Chapter 3.

1. 要求所有代数余子式之和，可求伴随矩阵

$adj.A = A^{-1} \cdot detA$

$$\begin{bmatrix} 0 & 2 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ -4 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\sim \begin{bmatrix} 0 & 1 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & \frac{1}{3} & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} \end{bmatrix}$$

✓ $\begin{bmatrix} 1&0&0&0&0&0&0&-\frac{1}{4} \\ 0&1&0&0&\frac{1}{2}&0&0&0 \\ 0&0&1&0&0&\frac{1}{3}&0&0 \\ 0&0&0&1&0&0&1&0 \end{bmatrix}$ 所以 $A^{-1}=\begin{bmatrix} 0&0&0&-\frac{1}{4} \\ \frac{1}{2}&0&0&0 \\ 0&\frac{1}{3}&0&0 \\ 0&0&1&0 \end{bmatrix}$

$\det A=\begin{vmatrix} 0&2&0&0 \\ 0&0&3&0 \\ 0&0&0&1 \\ -4&0&0&0 \end{vmatrix}=4\times\begin{vmatrix} 2&0&0 \\ 0&3&0 \\ 0&0&1 \end{vmatrix}$

$=4\times 6=24$

所以 $\text{adj}A=24A^{-1}=\begin{bmatrix} 0&0&0&-6 \\ 12&0&0&0 \\ 0&8&0&0 \\ 0&0&24&0 \end{bmatrix}$

代数余子式之和为 $-6+12+8+24=38$

2. (1) ①按第一行展开

$3\times\begin{vmatrix} 3&2 \\ 5&1 \end{vmatrix}+4\times\begin{vmatrix} 2&3 \\ 0&5 \end{vmatrix}$

$=3\times(3-10)+4\times(10-0)$

$=-21+40=19$

②按第二列展开

$3\times\begin{vmatrix} 3&4 \\ 0&1 \end{vmatrix}-5\times\begin{vmatrix} 3&4 \\ 2&2 \end{vmatrix}$

$=3\times(3-0)-5\times(6-8)$

$=9+10=19$

(2) ①按第一行展开

$$2\times\begin{vmatrix}0&3\\1&5\end{vmatrix}-3\times\begin{vmatrix}4&3\\6&5\end{vmatrix}-3\times\begin{vmatrix}4&0\\6&1\end{vmatrix}$$

$=2\times(0-3)-3\times(20-18)-3\times(4-0)$

$=-6-6-12=-24$

②按第二列展开

$$-3\times\begin{vmatrix}4&3\\6&5\end{vmatrix}+0\times\begin{vmatrix}2&-3\\6&5\end{vmatrix}-1\times\begin{vmatrix}2&-3\\4&3\end{vmatrix}$$

$=-3\times(20-18)+0-(6+12)$

$=-6-18=-24$

3.(1) $D=1\times\begin{vmatrix}3&3^2\\4&4^2\end{vmatrix}-1\times\begin{vmatrix}2&2^2\\4&4^2\end{vmatrix}+1\times\begin{vmatrix}2&2^2\\3&3^2\end{vmatrix}$

$=3\times4^2-3^2\times4-2\times4^2+2^2\times4+2\times3^2-2^2\times3$

$=(3-2)4^2+(2-4)3^2+(4-3)\times2^2$

$=4^2-2\times3^2+2^2=16-18+4=2$

(2) $D=1\times\begin{vmatrix}a_2&a_3\\a_2^2&a_3^2\end{vmatrix}-1\times\begin{vmatrix}a_1&a_3\\a_1^2&a_3^2\end{vmatrix}+1\times\begin{vmatrix}a_1&a_2\\a_1^2&a_2^2\end{vmatrix}$

$= a_2a_3^2 - a_2^2a_3 - a_1a_3^2 + a_1^2a_3 + a_1a_2^2 - a_1^2a_2$

~~$= a_3(a_1^2 - a_2^2) + a_2(a_3^2 - a_1^2) + a_1(a_2^2 - a_3^2)$~~

$= a_1a_2a_3 - a_1a_2a_3 + a_2a_3^2 - a_2^2a_3 - a_1a_3^2 + a_1^2a_3 + a_1a_2^2 - a_1^2a_2$

$= a_3(a_2a_3 - a_1a_2 - a_1a_3 + a_1^2)$

$\quad - a_2(a_2a_3 - a_1a_2 - a_1a_3 + a_1^2)$

$= (a_3 - a_2)(a_2a_3 - a_1a_2 - a_1a_3 + a_1^2)$

$= (a_3 - a_2)[a_2(a_3 - a_1) - a_1(a_3 - a_1)]$

$= (a_3 - a_2)(a_2 - a_1)(a_3 - a_1)$

4. 由(2)知. $D = (a_3 - a_2)(a_2 - a_1)(a_3 - a_1)$

猜测 $D_n = (a_2 - a_1)(a_3 - a_1)(a_3 - a_2)\cdots(a_n - a_1)\cdots(a_n - a_{n-1})$

$D_2 = \begin{vmatrix} 1 & 1 \\ a_1 & a_2 \end{vmatrix} = a_2 - a_1$.

假设 $D_{n-1} = (a_2 - a_1)(a_3 - a_1)(a_3 - a_2)\cdots(a_{n-1} - a_1)\cdots(a_{n-1} - a_{n-2})$ 成立.

$$\text{要 } D_n = \begin{vmatrix} 1 & 1 & 1 & \cdots & 1 \\ a_1 & a_2 & a_3 & \cdots & a_n \\ a_1^2 & a_2^2 & a_3^2 & \cdots & a_n^2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_1^{n-1} & a_2^{n-1} & a_3^{n-1} & \cdots & a_n^{n-1} \end{vmatrix}$$

对每一行减去上一行的 a_1 倍得

$$D_n=\begin{vmatrix} 1 & 1 & 1 & \cdots & 1 \\ 0 & a_2-a_1 & a_3-a_1 & \cdots & a_n-a_1 \\ 0 & a_2^2-a_2a_1 & a_3^2-a_3a_1 & \cdots & a_n^2-a_na_1 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & a_2^{n-1}-a_2^{n-2}a_1 & a_3^{n-1}-a_3^{n-2}a_1 & \cdots & a_n^{n-1}-a_n^{n-2}a_1 \end{vmatrix}$$

化简得.

$$D_n=\begin{vmatrix} 1 & 1 & 1 & \cdots & 1 \\ 0 & a_2-a_1 & a_3-a_1 & \cdots & a_n-a_1 \\ 0 & a_2(a_2-a_1) & a_3(a_3-a_1) & \cdots & a_n(a_n-a_1) \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & a_2^{n-2}(a_2-a_1) & a_3^{n-2}(a_3-a_1) & \cdots & a_n^{n-2}(a_n-a_1) \end{vmatrix}$$

展开得.

$$D_n=1\times\begin{vmatrix} a_2-a_1 & a_3-a_1 & \cdots & a_n-a_1 \\ a_2(a_2-a_1) & a_3(a_3-a_1) & \cdots & a_n(a_n-a_1) \\ \vdots & \vdots & & \vdots \\ a_2^{n-2}(a_2-a_1) & a_3^{n-2}(a_3-a_1) & \cdots & a_n^{n-2}(a_n-a_1) \end{vmatrix}$$

将矩阵转置提取因式后再转置为原顺序得.

$$D_n=(a_2-a_1)(a_3-a_1)\cdots(a_n-a_1)\times\begin{vmatrix} 1 & 1 & \cdots & 1 \\ a_2 & a_3 & \cdots & a_n \\ \vdots & \vdots & & \vdots \\ a_2^{n-2} & a_3^{n-2} & \cdots & a_n^{n-2} \end{vmatrix}$$

将此处矩阵按假设展开为

$(a_3-a_2)(a_4-a_2)(a_4-a_3)\cdots(a_n-a_2)\cdots(a_n-a_{n-1})$

则 $D_n=(a_2-a_1)(a_3-a_1)(a_3-a_2)\cdots(a_n-a_1)(a_n-a_2)\cdots(a_n-a_{n-1})$

猜想成立.

即由数学归纳法得.

$D_n=(a_2-a_1)(a_3-a_1)(a_3-a_2)\cdots(a_n-a_1)\cdots(a_n-a_{n-1})$

5.(1) $D=2\times3\times4+1\times0\times1+1\times0\times0-1\times3\times1-1\times1\times4-2\times0\times0$

$=24-3-4=17$.

(2) $D=a_1\begin{vmatrix} a_2 & 0 & 0 & 0 \\ 0 & a_3 & 0 & 0 \\ 0 & 0 & \ddots & \vdots \\ 0 & 0 & \cdots & a_n \end{vmatrix}+$

(2) $D=\begin{vmatrix} 1 & 1 & \cdots & 1 \\ a_2 & 0 & \cdots & 0 \\ 0 & a_3 & \cdots & 0 \\ \cdots & \cdots & \ddots & a_{n-1} \\ 0 & 0 & \cdots & a_n \end{vmatrix}$

(2) $D=a_2\begin{vmatrix} a_1 & 1 & 1 & \cdots & 1 \\ \frac{1}{a_2} & 1 & 0 & \cdots & 0 \\ 1 & 0 & a_3 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ 1 & 0 & 0 & \cdots & a_n \end{vmatrix} = a_2a_3\begin{vmatrix} a_1 & 1 & 1 & \cdots \\ \frac{1}{a_2} & 1 & 0 & \cdots \\ \frac{1}{a_3} & 0 & 1 & \cdots \\ \cdots & \cdots & \cdots & \cdots \\ 1 & 0 & 0 & \cdots a_n \end{vmatrix}$

$\cdots\cdots = a_2a_3a_4\cdots a_n\begin{vmatrix} a_1 & 1 & 1 & \cdots & 1 \\ \frac{1}{a_2} & 1 & 0 & \cdots & 0 \\ \frac{1}{a_3} & 0 & 1 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ \frac{1}{a_n} & 0 & 0 & \cdots & 1 \end{vmatrix}$

第二行及以后乘(-1)与第一行求和.

$D=a_2a_3a_4\cdots a_n\begin{vmatrix} a_1-\frac{1}{a_2}-\frac{1}{a_3}-\cdots\frac{1}{a_n} & 0 & 0 & \cdots 0 \\ \frac{1}{a_2} & 1 & 0 & \cdots 0 \\ \frac{1}{a_3} & 0 & 1 & \cdots 0 \\ \cdots & \cdots & \cdots & \cdots \\ \frac{1}{a_n} & 0 & 0 & \cdots \end{vmatrix}$

展开得

$a_2a_3a_4\cdots a_n\cdot(a_1-\frac{1}{a_2}-\frac{1}{a_3}-\cdots\frac{1}{a_n})\cdot\begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & 0 & 1 \end{vmatrix}$

$=(a_1-\frac{1}{a_2}-\frac{1}{a_3}-\cdots\frac{1}{a_n})\cdot a_2a_3a_4\cdots a_n$

$$\begin{vmatrix} a_1 & 1 & 1 & \cdots \\ a_2 & 1 & 0 & \cdots 0 \\ a_3 & 0 & 1 & \cdots 0 \\ \cdots & \cdots & \cdots & \cdots \\ 1 & 0 & 0 & \cdots a_n \end{vmatrix}$$

$$\begin{vmatrix} 1 & & & \\ & 0 & & \\ & & 0 & \\ & & \cdots & \\ & & & 1 \end{vmatrix}$$

$$\begin{vmatrix} 0 & \cdots & 0 \\ 0 & \cdots & 0 \\ 1 & \cdots & 0 \end{vmatrix}$$

6.(1) 每一列(除第一列)减去第一列得

$$\begin{vmatrix} 1+a_1 & -1 & 0 & \cdots & -1 \\ a_2 & 1 & 0 & \cdots & 0 \\ \cdots & \cdots & \cdots & & 0 \\ a_n & 0 & 0 & \cdots & 1 \end{vmatrix}$$

第二行及以后行与第一行相加得

$$\begin{vmatrix} 1+a_1+a_2+\cdots a_n & 0 & \cdots & 0 \\ a_2 & 1 & \cdots & \\ \cdots & \cdots & \cdots & \\ a_n & 0 & \cdots & 1 \end{vmatrix}$$

展开得 $(1+a_1+a_2+\cdots a_n)$ $\begin{vmatrix} 1 & 0 & \cdots & 0 \\ 0 & \cdots & 0 & \\ 0 & 0 & \cdots & 1 \end{vmatrix}$

$= 1+a_1+a_2+\cdots a_n$

(2) 每一列除第一列外减去第一列得.

$$\begin{vmatrix} x & y-x & y-x & y-x & y-x \\ y & x-y & 0 & 0 & 0 \\ y & 0 & x-y & 0 & 0 \\ y & 0 & 0 & x-y & 0 \\ y & 0 & 0 & 0 & x-y \end{vmatrix}$$

(3)

再每一行除第一行外加到第一行得.

$$\begin{vmatrix} x+4y & 0 & 0 & 0 & 0 \\ y & x-y & & & \\ y & & x-y & & \\ y & & & x-y & \\ y & & & & x-y \end{vmatrix}$$

$$= (x-y)^4 \begin{vmatrix} x+4y & 0 & 0 & 0 & 0 \\ y & 1 & 0 & 0 & 0 \\ y & 0 & 1 & 0 & 0 \\ y & 0 & 0 & 1 & 0 \\ y & 0 & 0 & 0 & 1 \end{vmatrix}$$

展开得 $(x-y)^4 \cdot (x+4y) \cdot \begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix}$

$= (x-y)^4 (x+4y)$

(3) x^5.

$$\begin{vmatrix} 1 & \frac{y}{x} & 0 & 0 & 0 \\ 0 & 1 & \frac{y}{x} & 0 & 0 \\ 0 & 0 & 1 & \frac{y}{x} & 0 \\ 0 & 0 & 0 & 1 & \frac{y}{x} \\ \frac{y}{x} & 0 & 0 & 0 & 1 \end{vmatrix}$$

令 $\frac{y}{x} = t$.

明. x^5

$$\begin{vmatrix} 1 & t & 0 & 0 & 0 \\ 0 & 1 & t & 0 & 0 \\ 0 & 0 & 1 & t & 0 \\ 0 & 0 & 0 & 1 & t \\ t & 0 & 0 & 0 & 1 \end{vmatrix}$$

展开得

$$
\begin{vmatrix} 1 & t & 0 & 0 \\ 0 & 1 & t & 0 \\ 0 & 0 & 1 & t \\ 0 & 0 & 0 & 1 \end{vmatrix}
\begin{matrix} x^5 \\ \\ \\ \\ \\ \end{matrix}
\begin{matrix} \\ \\ \\ \\ -x^5t \cdot \\ \end{matrix}
\begin{vmatrix} 0 & t & 0 & 0 \\ 0 & 1 & t & 0 \\ 0 & 0 & 1 & t \\ t & 0 & 0 & 1 \end{vmatrix}
$$

$$
= x^5 \begin{vmatrix} 1 & t & 0 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{vmatrix} - x^5 t \begin{vmatrix} 0 & t & 0 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{vmatrix}
$$

$$
+ x^5 t^2 \begin{vmatrix} t & 0 & 0 \\ 1 & t & 0 \\ 0 & 1 & t \end{vmatrix}
$$

$$
= x^5 + 0 + x^5 t^2 \cdot t^3
$$

$$
= x^5 + x^5 \cdot \frac{y^5}{x^5} = x^5 + y^5
$$

7. $D_n = \begin{vmatrix} a_1 & -1 & 0 & \cdots & 0 & 0 \\ a_2 & x & -1 & \cdots & 0 & 0 \\ a_3 & 0 & x & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{n-1} & 0 & 0 & \cdots & x & -1 \\ \frac{a_n}{x} & 0 & 0 & \cdots & 0 & 1 \end{vmatrix}$

最后一行加到前一行得

$x \neq 0$

$$\begin{vmatrix} a_1 & -1 & 0 & \cdots & 0 & 0 \\ a_2 & x & -1 & \cdots & 0 & 0 \\ a_3 & 0 & x & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{n-1}+\frac{a_n}{x} & 0 & 0 & \cdots & x & 0 \\ \frac{a_n}{x} & 0 & 0 & \cdots & 0 & 1 \end{vmatrix}$$

$\cdots$

$$D_n = x^{n-1}\begin{vmatrix} a_1+\frac{a_2}{x}+\cdots\frac{a_n}{x^{n-1}} & 0 & 0 & \cdots & 0 & 0 \\ a_2+\frac{a_3}{x}+\cdots\frac{a_n}{x^{n-2}} & 1 & 0 & \cdots & 0 & 0 \\ a_3+\frac{a_4}{x}+\cdots\frac{a_n}{x^{n-3}} & 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{n-1}+\frac{a_n}{x} & 0 & 0 & \cdots & 1 & 0 \\ \frac{a_n}{x} & 0 & 0 & \cdots & 0 & 1 \end{vmatrix}$$

按第一行展开得.

$$D_n = x^{n-1}\left(a_1+\frac{a_2}{x}+\cdots\frac{a_n}{x^{n-1}}\right)\begin{vmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & \vdots \\ 0 & 0 & \ddots & \vdots \\ \vdots & \vdots & & \\ 0 & 0 & \cdots & 1 \end{vmatrix}$$

$$= a_1x^{n-1}+a_2x^{n-2}+\cdots a_n$$

Cramer's Rule

1. $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 3 \\ 1 & 1 & -2 \end{bmatrix}$ $b = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$

$Ax = b$. $x_i = \dfrac{\det A(b_i)}{\det A}$

$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ $\det A = 4$

(Ans)

$$x_1 = \frac{\begin{vmatrix} 1 & 2 & 1 \\ 3 & 3 & 3 \\ 1 & 1 & -2 \end{vmatrix}}{4} = \frac{9}{4}$$

$$x_2 = \frac{\begin{vmatrix} 1 & 1 & 1 \\ 2 & 3 & 3 \\ 1 & 1 & -2 \end{vmatrix}}{4} = \frac{-3}{4}$$

$$x_3 = \frac{\begin{vmatrix} 1 & 2 & 1 \\ 2 & 3 & 3 \\ 1 & 1 & 1 \end{vmatrix}}{4} = \frac{1}{4}$$

2. 设 $A^{-1}=\begin{bmatrix} x_1 & x_4 & x_7 \\ x_2 & x_5 & x_8 \\ x_3 & x_6 & x_9 \end{bmatrix}$ $\det A=-7$

$$x_1=\frac{\begin{vmatrix} 1 & 1 & 2 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{vmatrix}}{\det A}=-\frac{1}{7}$$

$$x_2=\frac{\begin{vmatrix} 0 & 1 & 2 \\ 1 & 0 & 0 \\ 2 & 0 & 1 \end{vmatrix}}{\det A}=\frac{1}{7}$$

$$x_3=\frac{\begin{vmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 2 & -1 & 0 \end{vmatrix}}{\det A}=\frac{3}{7}$$

$$x_4=\frac{\begin{vmatrix} 0 & 1 & 2 \\ 1 & 1 & 0 \\ 0 & -1 & 1 \end{vmatrix}}{\det A}=\frac{3}{7}$$

$$x_5=\frac{\begin{vmatrix} 0 & 0 & 2 \\ 1 & 0 & 0 \\ 2 & 0 & 1 \end{vmatrix}}{\det A}=\frac{4}{7}$$

$$x_6=\frac{\begin{vmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \\ 2 & -1 & 0 \end{vmatrix}}{\det A}=-\frac{2}{7}$$

$$x_7=\frac{\begin{vmatrix}0&1&2\\0&1&0\\1&-1&1\end{vmatrix}}{\det A}=\frac{2}{7}\qquad x_8=\frac{\begin{vmatrix}0&0&2\\1&0&0\\2&1&1\end{vmatrix}}{\det A}=\frac{2}{7}$$

$$x_9=\frac{\begin{vmatrix}0&1&0\\1&1&0\\2&-1&1\end{vmatrix}}{\det A}=\frac{1}{7}$$

所以
$$A^{-1}=\begin{vmatrix}-\frac{1}{7}&\frac{3}{7}&\frac{2}{7}\\\frac{1}{7}&\frac{4}{7}&-\frac{2}{7}\\\frac{3}{7}&-\frac{2}{7}&\frac{1}{7}\end{vmatrix}$$

3. 设A为方程组系数矩阵

$$A=\begin{bmatrix}\lambda&1&1\\1&\lambda&1\\1&1&\lambda\end{bmatrix}$$

方程组有非零解，则A行列式$\det A=0$

$$\det A=\begin{vmatrix}\lambda&1&1\\1&\lambda&1\\1&1&\lambda\end{vmatrix}=0$$

则 $\lambda^3+1+1-\lambda-\lambda-\lambda=0$.

$\lambda^3-3\lambda+2=0$

即 $(\lambda-1)^2(\lambda+2)=0$

~~$\lambda\neq1$ 且 $\lambda\neq-2$~~

所以 $\lambda=1$ 或 $\lambda=-2$